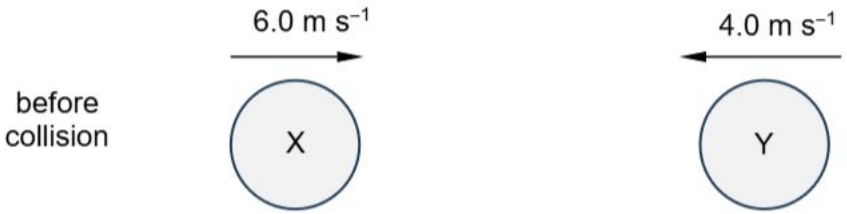
9 Body X of mass 2.5 kg undergoes a head-on collision with body Y of mass 1.0 kg. Bodies X and Y have initial speeds of 6.0 m s^{-1} and 4.0 m s^{-1} respectively.

The two bodies are in contact for a time of 20 ms. After the collision, body X continues in the same direction with speed 2.0 m s^{-1}.



What is the magnitude of the average force exerted on body Y during this collision?

A 10 N **B** 200 N **C** 250 N **D** 500 N

Ans: **D**

Change in momentum of X:

$\Delta p_x = m_x(v_x - u_x) = (2.5)(2.0 - 6.0) = -10.0$ N s

By N3L:

$\Delta p_y = -\Delta p_x = 10.0$ N s

$F_{avg}\Delta t = \Delta p_y = 10.0$ N s

$$F_{avg} = \frac{10}{20 \times 10^{-3}} = 500 \text{ N}$$

A: forgot to divide by time

B: used mass of Y

C: didn't calculate change in momentum, used final momentum of Y